DEXTROSE 10%

Hypoglycaemia treatment
Neonatal maintenance

CHEMICAL NAME Glucose 100 g/L in water — 555 mmol/L; ~400 kcal/L; hypertonic (~555 mOsm/L)
CLASS Hypertonic glucose solution
TRADE NAMES D10W · Glucose 10% infusion

THERAPEUTIC USES / INDICATIONS

- Treatment of hypoglycaemia (adults: 150–200 mL; children/neonates: 2 mL/kg) — safer than 50%.
- Neonatal maintenance fluids (with electrolytes).
- With insulin infusion in DKA when glucose <14 mmol/L.
- Hyperkalaemia regimen (with insulin); alternative to 50%; prevention of hypoglycaemia (sulfonylurea overdose, insulinoma).

ADMINISTRATION

- **IV** via peripheral line (large vein; irritant at high rates) or central.
- Infusion pump; frequent glucose monitoring.

DOSAGES

- **Adult hypoglycaemia:** 150–200 mL over 15 min (15–20 g glucose); recheck glucose in 10–15 min.
- **Child/neonate hypoglycaemia:** 2 mL/kg bolus, then infusion (neonates 60–90 mL/kg/day day 1).
- **DKA:** 125 mL/h alongside 0.9% saline when glucose <14 mmol/L.
- **Hyperkalaemia:** 250 mL 10% glucose + 10 units insulin over 15–30 min.

PHARMACOKINETICS

- Glucose metabolised; water distributed in total body water.
- **Onset:** glucose rises within minutes.

PHARMACODYNAMICS

- Supplies **glucose** for energy, prevents hypoglycaemia and ketosis; drives K^+ into cells with insulin.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Phlebitis, pain at site.
- Hyperglycaemia.

ADVERSE EFFECTS (SERIOUS)

- Hyponatraemia (free water) with large volumes.
- Rebound hypoglycaemia (hyperinsulinism) after stopping.
- Fluid overload; hypokalaemia; refeeding syndrome.

CONTRAINDICATIONS & CAUTIONS

- Hyperglycaemia; intracranial haemorrhage (relative).
- **Caution:** fluid overload, hyponatraemia.

KEY INTERACTIONS

- **Insulin:** combined for hyperkalaemia/DKA.
- **Thiamine:** give first in malnourished/alcohol misuse.

NURSING MANAGEMENT

- Capillary glucose 10–15 min after bolus, then hourly during infusion.
- Taper infusion slowly (rebound hypoglycaemia).
- Monitor Na^+, K^+ and fluid balance; check site.

PATIENT EDUCATION

- After recovering from a hypo, eat a starchy snack or meal.

KEY POINTS

- 10% glucose; preferred for IV hypo treatment.
- Adult 150–200 mL; child 2 mL/kg.
- DKA: add when glucose <14.
- Monitor glucose; taper slowly.

SOURCES CHECKED **[959]** Glucose 10% Intravenous Infusion (UK SmPC) **[448]** UKKA hyperkalaemia guideline (2023) **[625]** JBDS-IP hypoglycaemia guideline (2018) **[587]** JBDS-IP DKA guideline (2023) **[588]** Umpierrez et al., Diabetologia 2024
Full references p. 432–457